

Machine Learning Term Project

Project Title: Student Dropout Prediction

Name: Tayyaba Noor

Roll No: 33100

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1. Introduction

Student dropout is one of the major challenges faced by educational institutions. Students may leave their studies because of poor academic performance, financial difficulties, lack of motivation, family issues, or personal reasons. Predicting students who are at risk of dropping out allows institutions to provide guidance and support before students discontinue their education.

Machine learning makes it possible to analyze historical student data and identify patterns that indicate dropout risk. This project applies machine learning techniques to classify students as either likely to continue their education or likely to drop out.

2. Problem Statement

Educational institutions often struggle to identify students who may leave their studies before graduation. Manual identification is difficult and time-consuming. Therefore, there is a need for an automated machine learning system that predicts student dropout using historical student information.

3. Objectives

The objectives of this project are:

- To analyze student data.
- To preprocess and clean the dataset.
- To build a machine learning model for student dropout prediction.
- To evaluate model performance using different evaluation metrics.

- To predict whether a student is likely to drop out.

4. Dataset Description

The dataset contains information about students that can influence their academic performance and continuation in education.

Typical attributes include:

Martial Status

Application mode

Application order

Course

Daytime/evening attendance

Previous qualification

Previous qualification (grade)

Nationality

Mother's qualification

Father's qualification

Mother's occupation

Father's occupation

Admission grade

Displaced

Educational special needs

Debtor

Tuition fees up to date

Gender

Scholarship holder

Age at enrollment

International

Curricular units 1st sem (credited)

Curricular units 1st sem (enrolled)

Curricular units 1st sem (evaluations)

Curricular units 1st sem (approved)

Curricular units 1st sem (grade)

Curricular units 1st sem (without evaluations)

Curricular units 2nd sem (credited)

Curricular units 2nd sem (enrolled)

Curricular units 2nd sem (evaluations)

Curricular units 2nd sem (approved)

Curricular units 2nd sem (grade)

Curricular units 2nd sem (without evaluations)

Unemployment rate

Inflation rate

GDP

Target

The target variable indicates whether the student dropped out or continued their education.

5. Methodology

The following methodology was followed during the project:

1. Dataset collection
2. Data cleaning
3. Handling missing values
4. Encoding categorical variables
5. Feature selection
6. Splitting data into training and testing sets
7. Training machine learning models
8. Evaluating model performance
9. Making predictions

6. Data Preprocessing

Before training the machine learning model, the dataset was preprocessed.

The preprocessing steps included:

- Removing duplicate records

- Handling missing values
- Encoding categorical features
- Feature scaling (where required)
- Splitting data into training and testing datasets

Proper preprocessing improves the performance and accuracy of machine learning models.

7. Machine Learning Algorithms

Different supervised machine learning algorithms can be used for this project. Examples include:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

For this project, the selected algorithm was trained using the processed dataset to predict student dropout.

8. Model Training

The dataset was divided into training and testing sets. The training data was used to build the machine learning model, while the testing data was used to evaluate its performance.

The model learned patterns from student records and classified whether a student is likely to drop out.

9. Model Evaluation

The performance of the model was evaluated using standard classification metrics, including:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics help determine how effectively the model predicts student dropout.


```

import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix

```

```

from google.colab import files

```

```

uploaded = files.upload()

```

```

... Choose files data.csv
data.csv(text/csv) - 528805 bytes, last modified: 31/07/2026 - 100% done
Saving data.csv to data.csv

```

```

df = pd.read_csv("data.csv", sep=";")

# Remove spaces from column names if any
df.columns = df.columns.str.strip()

print(df.head())

```

```

      Marital status  Application mode  Application order  Course \
0                1                17                5      171
1                1                15                1     9254
2                1                 1                5     9878

```

```

print(df.columns.tolist())

```

```

... ['Marital status', 'Application mode', 'Application order', 'Course', 'Daytime/evening attendance', 'Previous qualification', 'Previous qualification']

```

```

print("Shape:", df.shape)

```

```

df.info()

```

```

Shape: (4424, 37)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4424 entries, 0 to 4423
Data columns (total 37 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Marital status                        4424 non-null  int64
1   Application mode                      4424 non-null  int64
2   Application order                    4424 non-null  int64
3   Course                              4424 non-null  int64
4   Daytime/evening attendance           4424 non-null  int64

```

```

plt.figure(figsize=(6,4))

```

```

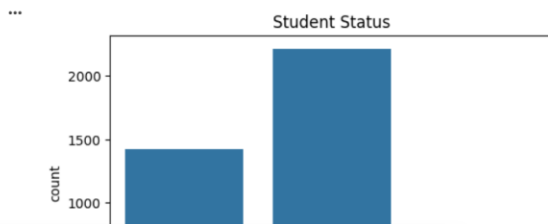
sns.countplot(data=df, x="Target")

```

```

plt.title("Student Status")
plt.show()

```



10. Results

The trained machine learning model successfully analyzed student information and predicted dropout status.

The evaluation results demonstrated that the model was able to classify students with good accuracy. The prediction system can assist educational institutions in identifying students who may require additional academic support.

	Model	Accuracy
0	Random Forest	0.767232
2	Logistic Regression	0.763842
1	Decision Tree	0.697175
3	KNN	0.577401

```
sample = X.iloc[[0]]
prediction = rf.predict(sample)
print("Prediction:", encoder.inverse_transform(prediction))
Prediction: ['Dropout']
```

11. Conclusion

This project successfully demonstrated the use of machine learning for predicting student dropout. Data preprocessing, feature engineering, and supervised learning techniques were applied to build a predictive model. The model can help educational institutions identify students who are at risk of dropping out and enable timely interventions. Such predictive systems can improve student retention and overall academic performance.